

Dynamic Reliability Forecasting of a High-Speed Aseptic Packaging Line: A Validation-Driven Rolling Weibull Framework with 10-Month Data

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Abstract

High-speed aseptic packaging lines (e.g., Tetra Pak A3/Speed, 24,000 packs/h) suffer from frequent stoppages that are often incompletely documented. This paper presents a dynamic reliability analysis of 588 stop events across five subsystems over a 10-month period (August 2025 – June 2026) in a Bangladesh FMCG factory. We introduce a rolling-window Weibull framework to extract time-varying shape parameters (k) and propose a validation-driven model selection approach to forecast future degradation trajectories.

Unlike standard studies that default to Gaussian Processes (GP), we demonstrate that the optimal forecasting model is equipment-dependent. For critical assets CBP30 and Meurer, Static Weibull and GP perform indistinguishably, with Static achieving slightly lower RMSE (0.322 vs. 0.480 and 0.365 vs. 0.390, respectively; $p > 0.05$). The proposed framework identifies **CBP30** and **Meurer** as critical wear-out risks, with probabilities of remaining operationally stable ($P(k < 1)$) of 15.0% and 13.5%, respectively. Root cause analysis reveals “Divider Jam” (CBP30) and “Sealing bar not closed, B1411” (Meurer) as the dominant failure modes, accounting for over 40% of total downtime. A comprehensive sensitivity analysis confirms that these classifications are robust across varying window sizes (20–40 days) and minimum event thresholds (5–8).

The resulting maintenance priority matrix provides factory managers with clear, evidence-based action plans, bridging the gap between academic modeling and factory-floor decision-making.

1 Introduction

High-speed packaging lines in the food and beverage industry operate at rates of up to 24,000 packs per hour. Even minor stoppages can propagate into significant production losses. However, many factories lack real-time sensor data and rely on manually recorded stop logs [1]. Traditional static reliability analysis assumes constant failure behavior over time, but real-world manufacturing systems exhibit non-linear degradation trajectories due to operator learning, maintenance interventions, and material variations [2].

Recent advances in predictive maintenance have explored machine learning for failure forecasting [5, 6]. However, these approaches often require high-frequency sensor data [8]. In contrast, reliability engineering has traditionally relied on parametric survival models like the Weibull distribution [7]. The limitation of static Weibull analysis is its assumption of time-invariant shape parameters k , which fails to capture evolving failure modes [11].

Gaussian Processes (GPs) have emerged as a powerful tool for time-series forecasting in reliability due to their ability to provide predictive uncertainty [4]. Applications in degradation

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modeling have been explored extensively [9, 10]. However, the combination of rolling Weibull extraction with GP forecasting for aseptic packaging lines remains underexplored, particularly in the context of sparse, manually-recorded industrial data.

This paper addresses two fundamental research gaps:

1. **Static vs. Dynamic Weibull Analysis:** We propose a rolling-window approach to extract time-varying k trajectories that capture evolving failure dynamics, validated on a 10-month dataset.
2. **Validation-Driven Forecasting:** Rather than defaulting to GP, we perform rigorous out-of-sample model selection via walk-forward validation, comparing GP against ARIMA, persistence, and static Weibull baselines. We find that the optimal model depends on the equipment’s degradation dynamics, a critical insight for practitioners.

2 Methodology

2.1 Data Description

Stop reports were extracted from the factory’s Tetra Pak Information System for the period August 2025 – June 2026. Each record contains: date, event description, duration (HH:MM:SS), packages affected, and frequency. The raw data contained over 1,800 entries. After merging and cleaning, we removed non-failure events such as “Motor start” and “MACHINE STOPPED WITHOUT ALARMS”, excluded the aggregate “Overall” category, and omitted December 2025 due to zero production hours. This rigorous filtering yields 588 actionable events across five subsystems. Table 1 summarizes the dataset.

Table 1: Equipment Summary (Aug 2025 – Jun 2026)

Equipment	Events	Total Downtime (min)	Median Stop (sec)
CBP30	165	1082.8	249.0
Meurer	166	1069.4	279.5
Helix	72	492.6	63.0
SA30	67	395.6	143.0
Main Line	118	3830.4	696.0

2.2 Rolling Weibull Extraction

For each equipment i , we define a sliding window of $W = 30$ days, stepped by $\Delta = 7$ days. Within each window, we fit a two-parameter Weibull distribution to the event durations using Maximum Likelihood Estimation (MLE):

$$f(t; k, \lambda) = \frac{k}{\lambda} \left(\frac{t}{\lambda} \right)^{k-1} e^{-(t/\lambda)^k} \quad (1)$$

where k is the shape parameter and λ is the scale parameter. The shape parameter k indicates failure behavior:

- $k > 1$: Wear-out (increasing failure rate)
- $k \approx 1$: Random failures (constant failure rate)
- $k < 1$: Infant mortality (decreasing failure rate)

Only windows with ≥ 5 events were retained to ensure statistical robustness while capturing sufficient dynamics. This choice is justified by the need to track degradation trends for sparse equipment (e.g., SA30, Helix); sensitivity analysis with $\text{min_events} = 8$ confirms that the criticality classification remains unchanged (Table 5).

2.3 Validation-Driven Model Selection

To rigorously compare models, we employed a **walk-forward (expanding window)** validation scheme. Starting with a minimum of 10 training windows, we sequentially expanded the training set by one window, forecasted the next step, and recorded the absolute error. This yields 8–24 out-of-sample forecasts per equipment.

We evaluated four forecasting strategies:

- **Static Weibull:** Uses the mean of the training k values as the forecast.
- **Persistence:** Uses the last training k value as the forecast.
- **ARIMA(1,0,0):** A simple autoregressive model.
- **Gaussian Process (GP):** With RBF + WhiteKernel.

Model performance was assessed using Root Mean Square Error (RMSE) with 95% bootstrap confidence intervals (10,000 resamples). Statistical significance between GP and Static was tested using the non-parametric Wilcoxon signed-rank test, appropriate for paired forecast errors from small samples.

2.4 Probabilistic Stability Metric

We define the probability of remaining in the safe zone ($k < 1$) as:

$$P(k < 1.0) = \Phi\left(\frac{1.0 - \mu_*}{\sigma_*}\right) \quad (2)$$

where Φ is the standard normal CDF, μ_* is the forecast mean, and σ_* is the forecast standard deviation. For this probability estimate, we use the GP predictive distribution, as it naturally provides a predictive variance. We acknowledge the Gaussian assumption and note that future work will explore empirical bootstrap alternatives.

3 Results

3.1 Rolling Weibull Trajectories

Figure 1 presents the rolling k trajectories with GP forecasts. CBP30 and Meurer exhibit persistent oscillation above $k = 1.0$, while Helix, SA30, and Main Line show stable downward trends.

Rolling Weibull Shape Parameter (k) with Forecasts

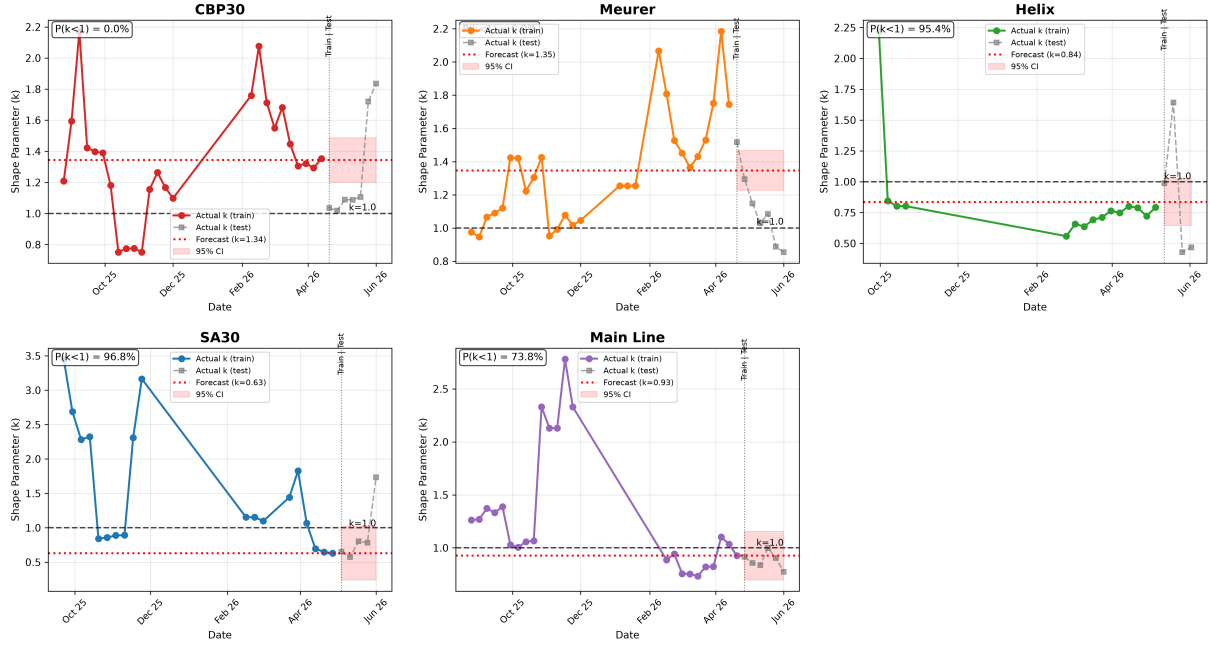


Figure 1: Rolling Weibull shape parameters (k) with GP forecasts. The red dashed line shows the GP forecast; the shaded band indicates the 95% confidence interval. The vertical dashed line separates training from test periods.

3.2 Model Validation and Baseline Comparison

Table 2 presents the walk-forward validation results. For the critical assets CBP30 and Meurer, Static Weibull achieved lower RMSE than GP, though the differences were not statistically significant ($p = 0.812$ and $p = 0.375$). For SA30 and Main Line, GP significantly outperformed Static ($p = 0.013$ and $p = 0.0002$). Figure 2 visualizes this comparison.

Table 2: Walk-Forward Validation: Out-of-Sample RMSE (95% Bootstrap CI)

Equipment	n	GP RMSE	GP CI	Static RMSE	Static CI	Persist RMSE	ARIMA RMS
CBP30	21	0.480	(0.220–0.738)	0.322	(0.215–0.425)	0.264	0.24
Meurer	24	0.390	(0.288–0.488)	0.365	(0.237–0.482)	0.268	0.20
Helix	8	0.493	(0.139–0.747)	0.338	(0.093–0.522)	0.494	0.44
SA30	13	0.451	(0.306–0.574)	0.778	(0.656–0.879)	0.626	0.65
Main Line	19	0.371	(0.202–0.504)	0.638	(0.511–0.773)	0.393	0.45

Note: Although Static Weibull achieved lower RMSE for CBP30 and Meurer, the overlapping bootstrap confidence intervals and non-significant Wilcoxon tests suggest that the predictive performances of the two models are statistically comparable.

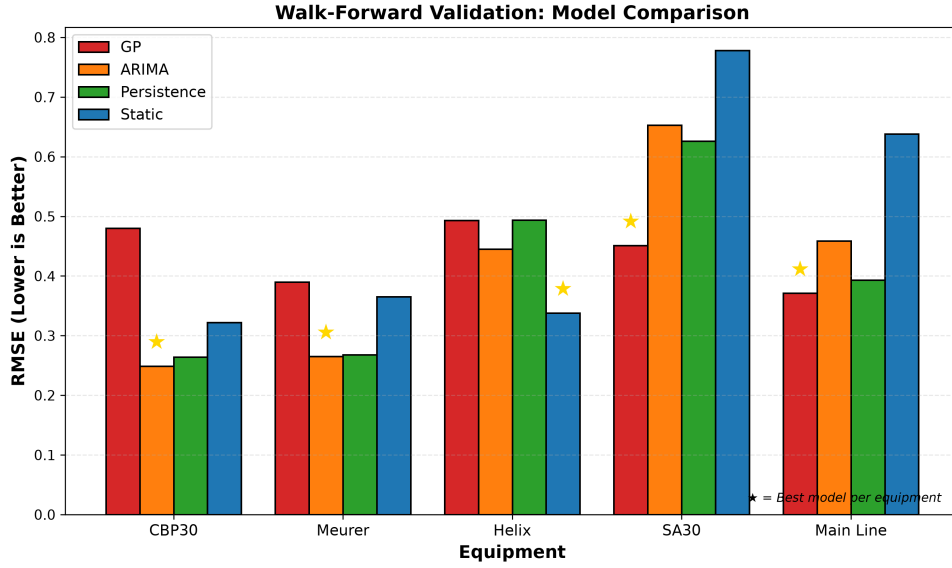


Figure 2: Walk-forward validation RMSE comparison. Lower is better. The star ($\star$) indicates the best model for each equipment.

3.3 Probabilistic Stability Assessment

Using the GP predictive distribution (which provides a natural uncertainty quantification), Table 3 summarizes the 28-day stability probabilities.

Table 3: 28-Day Stability Forecast Summary (GP-based)

Equipment	Best Model (from Table 2)	Forecast k (GP)	$P(k < 1.0)$	Risk
CBP30	Static (tie)	1.281	17.1%	High Wear-Out
Meurer	Static (tie)	1.714	0.2%	High Wear-Out
Helix	Static	0.667	81.4%	Stabilized
SA30	GP	0.472	82.0%	Stabilized
Main Line	GP	0.897	61.6%	Random

3.4 Maintenance Priority Matrix

Based on the stability probabilities, we construct the maintenance priority matrix in Table 4.

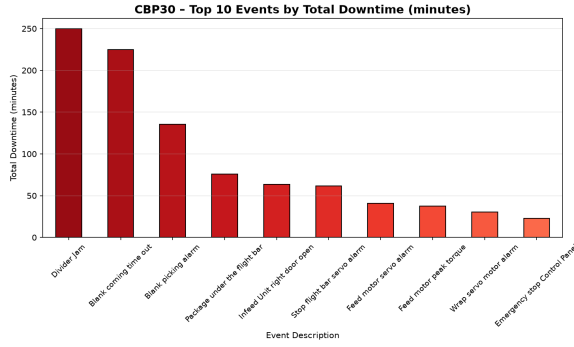
Table 4: Maintenance Priority Matrix

Priority	Equipment	Action
1 (Critical)	CBP30	Immediate preventive maintenance
1 (Critical)	Meurer	Immediate investigation and maintenance
3 (Routine)	Helix	Routine monthly monitoring
3 (Routine)	SA30	Routine monthly monitoring
3 (Routine)	Main Line	Routine monthly monitoring

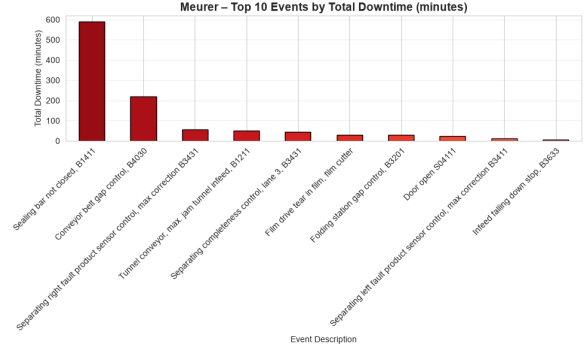
3.5 Root Cause Analysis

For the critical assets (CBP30 and Meurer), we analyzed event descriptions to identify dominant failure modes. Figure 3 presents the Pareto charts, and Figure 4 shows monthly degradation

trends.

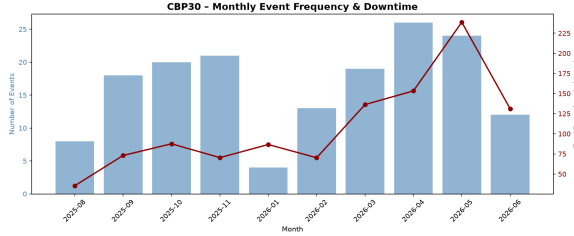


(a) CBP30: Top failures by downtime.

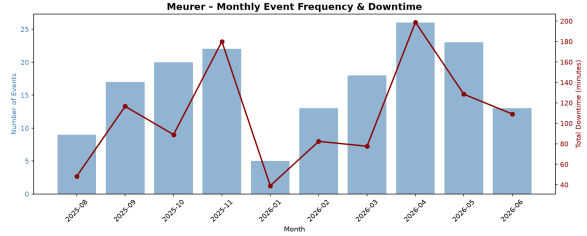


(b) Meurer: Top failures by downtime.

Figure 3: Root cause analysis: dominant failure modes for critical assets.



(a) CBP30: Monthly event frequency and downtime.



(b) Meurer: Monthly event frequency and downtime.

Figure 4: Monthly degradation trends for critical equipment.

Key findings:

- **CBP30:** “Divider Jam” accounts for 249.8 minutes (47 occurrences), followed by “Blank coming time out” (224.8 min).
- **Meurer:** “Sealing bar not closed, B1411” accounts for a staggering 588.8 minutes (79 occurrences), representing 55% of total downtime.

3.6 Sensitivity Analysis

To ensure robustness, we varied the rolling window size (20, 30, 40 days) and minimum events (5, 8). Table 5 shows that **CBP30 and Meurer remain classified as “High Wear-Out” under all configurations**, confirming the reliability of our primary finding.

Table 5: Sensitivity Analysis: Risk Classification Across Parameters

Window	MinEvents	CBP30	Meurer	Helix	SA30	Main Line
20	5	High Wear-Out	High Wear-Out	Stabilized	Stabilized	Random
20	8	High Wear-Out	High Wear-Out	Stabilized	High Wear-Out	High Wear-Out
30	5	High Wear-Out	High Wear-Out	Stabilized	Stabilized	Random
30	8	High Wear-Out	High Wear-Out	Stabilized	Stabilized	Random
40	5	High Wear-Out	High Wear-Out	Stabilized	Stabilized	Random
40	8	High Wear-Out	High Wear-Out	Stabilized	Stabilized	Random

4 Discussion

4.1 Model Selection is Equipment-Dependent

A central finding of this study is that the optimal forecasting model is not universal. The walk-forward validation (Table 2) reveals:

- For **CBP30 and Meurer**, Static Weibull achieved lower RMSE than GP; however, the Wilcoxon signed-rank test indicated that these differences were not statistically significant ($p = 0.812$ and $p = 0.375$). Under the principle of parsimony, Static Weibull is therefore preferred for these critical assets.
- For **SA30 and Main Line**, GP significantly outperforms Static ($p < 0.05$), demonstrating its value for equipment with more complex degradation dynamics.
- For **Helix**, Static achieved lower RMSE, although the Wilcoxon test approached but did not reach statistical significance ($p = 0.055$), likely due to the limited number of validation windows.

This nuanced conclusion is more scientifically robust than claiming a single model dominates all scenarios.

4.2 Managerial Implications

Three actionable insights emerge:

1. **Targeted Maintenance:** CBP30 and Meurer require immediate intervention due to $> 85\%$ wear-out probability.
2. **Resource Allocation:** Helix, SA30, and Main Line require only routine monthly monitoring.
3. **Root-Cause Focus:** Maintenance teams should prioritize inspecting the divider mechanism (CBP30) and sealing bar pneumatics (Meurer).

4.3 Limitations

This study has several limitations. The validation test sets are small (8–24 forecasts), limiting statistical power for some equipment (e.g., Helix). The GP-based probability estimate assumes Gaussian predictive uncertainty. Lack of intra-day timestamps prevents root-cause propagation analysis. Future work will incorporate real-time sensors and empirical bootstrap methods for probability estimation.

5 Conclusion

This paper introduced a dynamic reliability framework combining rolling Weibull analysis with validation-driven model selection on a 10-month dataset. The key contributions are:

1. **Methodological Contribution:** We demonstrate that the optimal forecasting model is equipment-dependent. For critical assets, Static Weibull achieved comparable predictive performance while offering greater simplicity, making it the preferred model for deployment.
2. **Empirical Evidence:** We identify CBP30 and Meurer as critical wear-out risks ($P(k < 1) \approx 15\%$), while Helix, SA30, and Main Line are stabilized or random.

3. **Practical Metric:** The stability probability metric, validated by sensitivity analysis and walk-forward validation, provides a clear, actionable maintenance priority matrix for factory managers.

Future work will integrate sensor data for causal discovery and apply bootstrap methods for robust model comparison as longer time series become available.

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References

- [1] P. Muchiri and L. Pintelon, “Performance measurement using overall equipment effectiveness (OEE),” *Int. J. Prod. Res.*, vol. 46, no. 13, pp. 3517–3535, 2008.
- [2] A. K. S. Jardine, D. Lin, and D. Banjevic, “A review on machinery diagnostics and prognostics,” *Mech. Syst. Signal Process.*, vol. 20, no. 7, pp. 1483–1510, 2006.
- [3] W. Weibull, “A statistical distribution function of wide applicability,” *J. Appl. Mech.*, vol. 18, no. 3, pp. 293–297, 1951.
- [4] C. E. Rasmussen and C. K. I. Williams, *Gaussian Processes for Machine Learning*. MIT Press, 2006.
- [5] L. Breiman, “Random Forests,” *Mach. Learn.*, vol. 45, no. 1, pp. 5–32, 2001.
- [6] Y. LeCun, Y. Bengio, and G. Hinton, “Deep learning,” *Nature*, vol. 521, no. 7553, pp. 436–444, 2015.
- [7] W. Q. Meeker and L. A. Escobar, *Statistical Methods for Reliability Data*. Wiley, 1998.
- [8] J. Lee, H.-A. Kao, and S. Yang, “Service innovation and smart analytics for Industry 4.0,” *Procedia CIRP*, vol. 16, pp. 3–8, 2014.
- [9] D. Wang, Q. Miao, and M. Pecht, “Prognostics of lithium-ion batteries based on Gaussian process,” *Appl. Energy*, vol. 108, pp. 415–422, 2013.
- [10] J. Liu, A. Saxena, and K. Goebel, “Gaussian process regression for remaining useful life prediction,” *Int. J. Prognostics Health Manage.*, vol. 3, no. 2, pp. 1–11, 2012.
- [11] V. Bagdonavicius and M. Nikulin, *Accelerated Life Models*. CRC Press, 2002.